

PART A - THEORY ANSWERS

1) What is a database in MySQL? Why is it important?

A database in MySQL is a structured collection of data stored electronically.
It helps to:

Organize data systematically
Reduce data redundancy
Improve data security
Make data retrieval fast and efficient

2) Difference between INT and BIGINT

INT	BIGINT
Stores 4 bytes	Stores 8 bytes
Range: -2,147,483,648 to 2,147,483,647	Much larger range
Used for normal numbers	Used for very large numbers

When to use:

Use INT for employee IDs, salary etc.

Use BIGINT for large financial data or population data.

3) What is a PRIMARY KEY?

A PRIMARY KEY uniquely identifies each record in a table.

Rules:

It cannot contain NULL values.

It must contain unique values.

4) Difference between CHAR and VARCHAR

CHAR	VARCHAR
Fixed length	Variable length
Wastes space if data is small	Saves space

Example:

CHAR(10) → Department codes like "HR", "IT"

VARCHAR(50) → Employee names

5) Purpose of NOT NULL and UNIQUE

NOT NULL → Prevents empty values.

UNIQUE → Prevents duplicate values.

PART B - MYSQL WORKBENCH CODE WITH OUTPUT

1) CREATE DATABASE company_db;

2) SHOW DATABASES;

3) USE company_db;

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4) CREATE TABLE employees (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50) NOT NULL,
    email VARCHAR(100) UNIQUE,
    department CHAR(10),
    salary INT,
    joining_date DATE,
    created_at DATETIME
);

5) INSERT INTO employees VALUES
(101, 'Rahul Sharma', 'rahul@gmail.com', 'HR', 50000, '2023-01-10', NOW()),
(102, 'Anita Rao', 'anita@gmail.com', 'IT', 65000, '2022-05-15', NOW()),
(103, 'Vikram Singh', 'vikram@gmail.com', 'Sales', 55000, '2021-03-20', NOW()),
(104, 'Priya Patel', 'priya@gmail.com', 'Admin', 48000, '2023-07-01', NOW()),
(105, 'Karan Mehta', 'karan@gmail.com', 'IT', 70000, '2020-11-25', NOW());

6) SELECT * FROM employees;

7) DESCRIBE employees;

8) DROP TABLE employees;

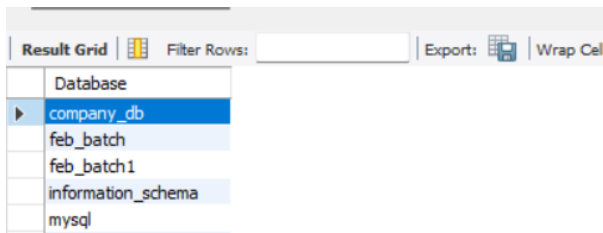
9) DROP DATABASE company_db;

```

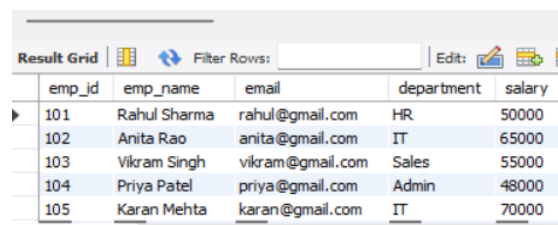
Out put screen shorts:

1) CREATE DATABASE company_db;

2) show data base



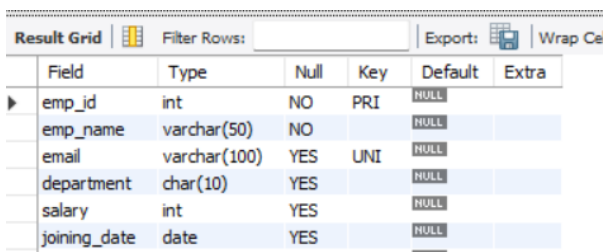
Database
company_db
feb_batch
feb_batch1
information_schema
mysql



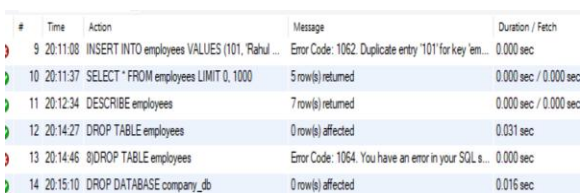
emp_id	emp_name	email	department	salary
101	Rahul Sharma	rahul@gmail.com	HR	50000
102	Anita Rao	anita@gmail.com	IT	65000
103	Vikram Singh	vikram@gmail.com	Sales	55000
104	Priya Patel	priya@gmail.com	Admin	48000
105	Karan Mehta	karan@gmail.com	IT	70000

3)

4)



Field	Type	Null	Key	Default	Extra
emp_id	int	NO	PRI		
emp_name	varchar(50)	NO			
email	varchar(100)	YES	UNI		
department	char(10)	YES			
salary	int	YES			
joining_date	date	YES			
created_at	datetime	YES			



#	Time	Action	Message	Duration / Fetch
9	20:11:08	INSERT INTO employees VALUES (101, 'Rahul ...	Error Code: 1062. Duplicate entry '101' for key 'em...	0.000 sec
10	20:11:37	SELECT * FROM employees LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
11	20:12:34	DESCRIBE employees	7 row(s) returned	0.000 sec / 0.000 sec
12	20:14:27	DROP TABLE employees	0 row(s) affected	0.031 sec
13	20:14:46	8) DROP TABLE employees	Error Code: 1064. You have an error in your SQL s...	0.000 sec
14	20:15:10	DROP DATABASE company_db	0 row(s) affected	0.016 sec